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Math 544/CptS. 531 - Fall 2024

Take home midterm exam

Work on your own. You may consult textbooks and lecture notes. To receive full credit, explain your steps and display your computations explicitly, within reason. You may use Matlab for Problem 1 (c). Cite any help you obtained from a chatbot.

1. Let $A = \begin{bmatrix} 2 & -1 & 1 \\ 6 & 1 & 2 \\ 2 & -1 & 3 \end{bmatrix}$ and $b = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}$. **[30 pts]**

   (a) Compute a lower triangular matrix L and upper triangular matrix U such that $A = LU$.

   (b) Make use of L and U to solve the system $Ax = b$.

   (c) Consider the iterative Jacobi method to solve the same system. Does it work? If not, think of preconditioning the system so that the Jacobi method works and perform 4 steps of the iteration. (Hint: Consider a permutation of the rows of A.)

2. Find a least-squares solution of the system $Ax = b$, where $A = \begin{bmatrix} 1 & -1 \\ 1 & -1 \\ 1 & 0 \end{bmatrix}$, $b = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$. Use any method you like. **[30 pts]**

3. Consider a subspace W of $\mathbb{R}^n$. **[40 pts]**

   (a) Give a precise definition of the orthogonal complement, $W^\perp$, of W.

   (b) Prove that $W^\perp$ is a subspace of $\mathbb{R}^n$.

   (c) Find an orthonormal basis of $W = \text{span}\{x, y\}$, where $x = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, $y = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}$.

   (d) Find the orthogonal projection of $z = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ onto the subspace W in (c).